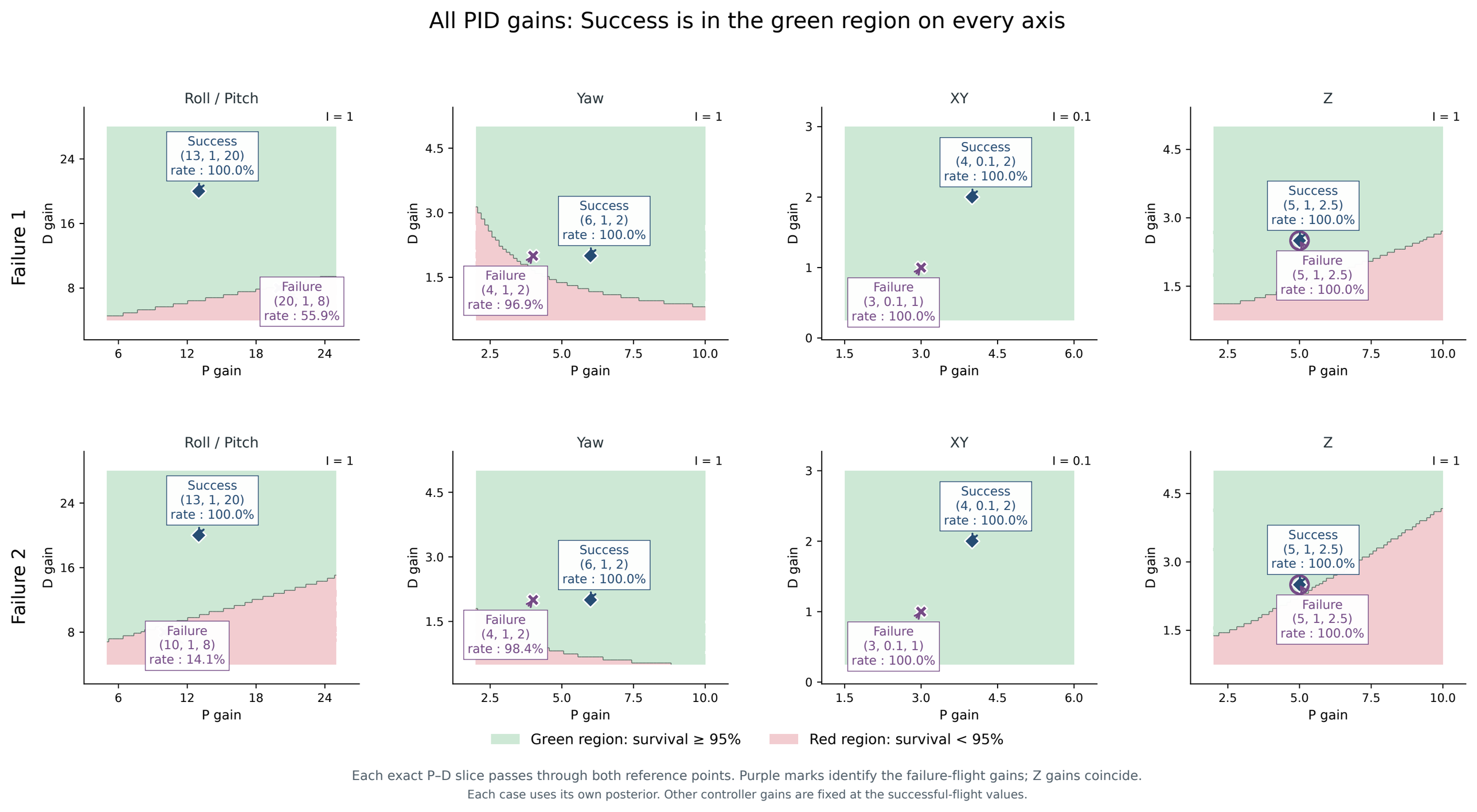
All PID gains: Success is in the green region on every axis
Failure 1
Failure 2
Roll / Pitch
Yaw
XY
Z
I = 1
I = 0.1
D gain
P gain
Success (13, 1, 20) rate : 100.0%
Failure (20, 1, 8) rate : 55.9%
Success (6, 1, 2) rate : 100.0%
Failure (4, 1, 2) rate : 96.9%
Success (4, 0.1, 2) rate : 100.0%
Failure (3, 0.1, 1) rate : 100.0%
Success (5, 1, 2.5) rate : 100.0%
Failure (5, 1, 2.5) rate : 100.0%
Failure (10, 1, 8) rate : 14.1%
Failure (4, 1, 2) rate : 98.4%
Green region: survival ≥ 95%
Red region: survival < 95%
Each exact P–D slice passes through both reference points. Purple marks identify the failure-flight gains; Z gains coincide.
Each case uses its own posterior. Other controller gains are fixed at the successful-flight values.